



1. What is the value of $20 \cdot 202020(2! + 0! + 2!) + 0!$?

- (A) 200 (B) 202 (C) 400 (D) 800 (E) 2020

2. In square $ABCD$, point E is the midpoint of segment AB , and point F is the midpoint of segment EB . If $BC = 12$, what is the length of DF ?

- (A) 14 (B) 15 (C) 16 (D) 18 (E) 20

3. Each student in a science club has a favorite subject, which is one of biology, chemistry, or physics. 25% of students have biology as their favorite subject. The number of students whose favorite subject is chemistry is 20% less than the number of students whose favorite subject is physics. 64 students in the club have either biology or physics as their favorite subject. How many students are in the club?

- (A) 84 (B) 90 (C) 96 (D) 102 (E) 108

4. Angle X is supplementary to angle Y , and angle Y is complementary to angle Z . Given that angles X , Y , and Z are all positive angles less than 180° , how many of the following angles must be acute?

The supplement of $\angle X$

The complement of $\angle Y$

The supplement of $\angle Y$

The complement of $\angle Z$

The supplement of $\angle Z$

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

5. Bonnie has a 25-gallon bucket which she wants to fill up with water. Starting at 12:00 PM, and after every five minutes thereafter, she pours one gallon of water on the bucket. However, there is a hole located in the bottom of the bucket which drains water out at a constant rate of 2.4 gallons per hour. At what time will the bucket first be completely filled with water?

- (A) 2:30 PM (B) 2:45 PM (C) 3:00 PM (D) 3:15 PM (E) 3:30 PM

6. Jeff is making Jane guess a "secret number", which is a positive integer. Jane knows the following:

The number has 12 positive divisors

The sum of the digits of the number is 12

The number is divisible by three primes whose sum is 12

What is the product of the digits in Jeff's secret number?

- (A) 0 (B) 16 (C) 24 (D) 32 (E) 36

7. Preston wants to buy k cupcakes for his friends. He has the choice to buy from at least one of two different stores - the first one sells strawberry cupcakes in groups of 6, while the second sells chocolate cupcakes in groups of 10. If Preston buys s strawberry cupcakes and c chocolate cupcakes, then there are four possible ordered pairs (s, c) , where s and c are nonnegative integers. What is the smallest possible value of k ?

- (A) 60 (B) 72 (C) 90 (D) 108 (E) 120

8. Given that $2^n - 1$ is a prime, which can be n below?

- (A) 44493 (B) 44494 (C) 44495 (D) 44496 (E) 44497

9. Suppose in a list of seven positive integers, the mean, median, and unique mode of the list are 1, 16, and 27 (not necessarily in that order). If M is the largest among the seven positive integers, what is the smallest possible value of M ?

- (A) 27 (B) 53 (C) 54 (D) 80 (E) 81

10. There are 6 people standing in a row. Ed, the first person from the left, is 149 cm tall. Al, the second person in the row, is 197 cm tall. For each of the other five people,

their height is the average of the heights of two distinct people to their left (which may not necessarily be the two people directly to their left). Which of the following can be the height of the sixth person in the row?

- (A) 150cm (B) 160cm (C) 170cm (D) 180cm (E) 190cm

11. If x is a real number such that $12x + \frac{25}{x} = 35$, what is the absolute value of $12x - \frac{25}{x}$?

- (A) 5 (B) 10 (C) 15 (D) 20 (E) 25

12. How many trailing zeroes are there when the number

$$\sqrt[20]{20^{2020} \cdot 2020^{20}}$$

is written out in base 2? (Trailing zeroes are consecutive zeroes at the end of an integer that do not contain any nonzero digits.)

- (A) 200 (B) 201 (C) 202 (D) 204 (E) 2020

13. Nina draws n lines on an infinite chalkboard, such that no two are parallel and no three concur. There are exactly x points in which two lines intersect, and the lines divide the board into y distinct regions. Which of the following is a possible value of $x + y$?

- (A) 130 (B) 140 (C) 150 (D) 160 (E) 170

14. Points E and F are located inside rectangle $ABCD$ such that $AE = EF = FC = 1$, $AE \perp EF$, $EF \perp FC$, and $BC = \sqrt{3}$. What is the area of rectangle $ABCD$?

- (A) $\sqrt{3}$ (B) 2 (C) $\sqrt{5}$ (D) $\sqrt{6}$ (E) $2\sqrt{3}$

15. The following rule is applied to check if a number is divisible by 59: take the last two digits, multiply the obtained two-digit number by n , then add the result to the number formed by the remaining digits. If the resulting number is divisible by 59, then the original number must also be divisible by 59. Given that n is positive, what is the remainder when it is divided by 59?

- (A) 3 (B) 15 (C) 21 (D) 36 (E) 42

16. If a, b, c, d are real numbers that satisfy the system

$$\begin{aligned}ab + ac + ad &= 135 \\bc + bd + cd &= 43 \\b^2 + c^2 + d^2 &= 139\end{aligned}$$

what is the value of a^2 ?

- (A) 72 (B) 81 (C) 90 (D) 99 (E) 108

17. Consider the real numbers $x = \sqrt{7 - 4\sqrt{3}} + \sqrt{7 + 4\sqrt{3}}$ and $y = \sqrt{3 - 2\sqrt{2}} + \sqrt{3 + 2\sqrt{2}}$. If $x^{147} = y^k$ for some positive integer k , what is k ?

- (A) 128 (B) 140 (C) 168 (D) 182 (E) 196

18. A right square pyramid whose base has side sm and whose height is hm has volume $11.2m^3$ and lateral surface area $23.2m^2$. (The lateral surface area is the total surface area of the pyramid excluding the base.)

Selim sees one of the triangular faces of the pyramid, which to his view is an isosceles triangle with base s and height h . This face has a small square painted on it. The area of the square is $174cm^2$, but in Selim's view, it is a trapezoid with area $126cm^2$. What is the value of $\lfloor s \rfloor$? ($\lfloor x \rfloor$ refers to the greatest integer less than or equal to x .)

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

19. What is the sum of the real roots of $x^4 - 3x^3 - 2x^2 - 3x + 1 = 0$?

- (A) -1 (B) 1 (C) 2 (D) 3 (E) 4

20. Let $x \spadesuit y$ denote the sum of the arithmetic mean and geometric mean of x and y . For example,

$$3 \spadesuit 27 = 3 + 27/2 + \sqrt{3 \cdot 27} = 15 + 9 = 24$$

If a , b and c are positive real numbers such that $a \spadesuit b = 50$, $b \spadesuit c = 32$, and $a \spadesuit c = 98$, what is $\sqrt{abc}$?

- (A) 96 (B) 128 (C) 140 (D) 156 (E) 180

21. For a positive integer p , call q a p -terminator if both pq and qp are rational numbers whose decimal representations are finite. For example, 3 is not a 5-terminator, as $53 = 1.\overline{6}$, which does not terminate. How many three-digit positive integers k are there such that k is a 650-terminator?

- (A) 3 (B) 6 (C) 9 (D) 12 (E) 15

22. In the coordinate plane with origin O , a line with slope $s > 1$ intersects the x-axis and y-axis at points X_1 and Y_1 , respectively, while a line with slope $1s$ intersects the x-axis and y-axis at points X_2 and Y_2 , respectively. The two lines intersect at point P , which is located in quadrant II. If $PX_1 = 6$, $PY_1 = 32$, $PX_2 = 8$, and $PY_2 = 24$, the distance OP is equal to $m\sqrt{np}$, where m , n , and p are positive integers, and n is not divisible by the square of a prime. What is $m + n + p$?

- (A) 46 (B) 47 (C) 48 (D) 49 (E) 50

23. In a regular octagon, Ozzy draws in every diagonal (i.e. a segment whose endpoints are non-consecutive vertices of the octagon). He then assigns to each diagonal a friendship score as follows: consider all intersections of two or more diagonals inside the octagon. If two diagonals intersect at a point, he gives both diagonals one point. If three diagonals concur at a point, he gives each diagonal two points (the point for two intersecting diagonals is not given), and if four diagonals concur at a point, he gives all four diagonals three points each (again, the points for two and three diagonals intersecting are not given). Ozzy then randomly selects a diagonal. What is the expected value of its friendship score?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

24. In acute triangle $\triangle ABC$ with $AB = 28$ and $AC = 32$, A_0 is the foot of the altitude from A to $\overline{BC}$, B_0 is the foot of the altitude from A_0 to $\overline{AC}$, and C_0 is the foot of the altitude from B_0 to AB . Segments AA_0 and B_0C_0 intersect at W . Given that $WB_0 = 7\sqrt{3}$ and $AB_0 > B_0C_0$, what is the area of $\triangle ABC$?

- (A) 168 (B) 240 (C) $160\sqrt{3}$ (D) $168\sqrt{3}$ (E) 336

25. A box contains 2020 balls, where each ball is colored one of red, yellow, or blue, and there is at least one ball of each color. Over half of the balls are red. Bradley takes out balls from the box, one at a time and without replacement, until he draws out a blue ball, at which point he stops. The expected number of yellow balls he has taken out is 21, while the expected number of red balls he has taken out is k , which is a positive integer. What is k ?

(A) 22 (B) 23 (C) 24 (D) 25 (E) 26

Time limit: 75 minutes